Leveraging IoT Wearable Technology Towards Early Diagnosis of Neurological Diseases

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***Abstract*— The leading trends in the framework of the Internet of Things are driving the research community to provide smart systems and solutions aimed at revolutionizing medical sciences and healthcare. One of the major opportunities offered by IoT lies in the ubiquitous connectivity, thus enabling smart services such as remote patient monitoring, in-home therapy/rehabilitation, and assisted living platforms. In this paper we present a prototype of wearable smart glasses able to monitor the Eye Blinks (EBs) through ElectroOculoGram (EOG) signal in a transparent way with respect to the final user. We propose a novel pre-filtering scheme to reduce EOG noise along with an analytical derivation of a matched filter to detect and count EBs. We have carried out an in-depth experimental campaign in order to validate the robustness of our approach with respect to the main solutions available in the literature. Furthermore, we have compared the performances obtained with out wearable prototype versus the results achievable with professional medical equipments. Results show that our solution is able to achieve very high accuracy in EB detection, obtaining comparable performance with respect to professional medical desktop equipment, with the additional benefit of portability, comfort and easiness of use for the patients.**

***Index Terms*— IoT, eHealth, wearable technology, eye blink, glasses, neurological diseases, signal processing.**

I. Introduction

INTERNET of Things (IoT) has emerged as one of the most trending topics of technology in recent times since it paves the way for smart and innovative applications and services suitable for a number of different scenarios [1]. Among other aspects, many IoT devices and applications aim at developing eHealth solutions, thus enabling the healthcare system with new services. The general purpose of IoT is that of improving the quality of life, by leveraging ubiquitous connectivity for millions of devices and systems [2]. The eHealth framework is certainly one of the most critical scenarios in which IoT and mobile sensor networks can play a major role [3], [4]. In this regard, the new technologies can revolutionize the healthcare system by introduction the Internet of Health Things (IoHT) [5] which enable new solutions such as remote patient monitoring, in-home therapy, block-chains [6], edge and cloud computing for eHealth [7], [8]. Provided the benefit that the general framework of healthcare will receive from leveraging ubiquitous connectivity and technologies, great focus has been put in finding specific smart solutions to detect, diagnose and cure common as well as delicate medical diseases [9]. Alzheimer's and Parkinson's diseases are among the most well-known neurological illnesses. According to the World Health Organization, in 2019 about 7 million people worldwide were affected by Alzheimer's or Parkinson's diseases [10]. Consequently, paramount importance is given to finding a novel and practical solution for the early detection of the related neurological symptoms. According to the medical literature, there is a strong correlation among the onset of these neurological illnesses and external symptoms such as the type of eye blinking activity. Eye Blink (EB) is a very rapid involuntary or voluntary movement that consists in closing and opening the eyes. Recent works in medical literature (see [11], [12] and references therein), stress the importance of monitoring EBs during fixation in Alzheimer's and Parkinson's diseases [13], [14]. Given the notable importance of early detection of symptoms, many methods have been proposed in literature to monitor eye blinking activity. The state of the art includes different approaches to the problem: some existing solutions employ medical fixed equipment, while others rely on the use of wearable IoT technology. To maximize the probability of diagnosing neurological diseases in their very early stages it is important to provide real-time and continuous monitoring of the external symptoms. This can only be performed by employing non-invasive, patient-friendly, wearable devices.

In this paper we present a prototype of sensored glasses able to count the number of EBs in a transparent way with respect to the final user. In order to perform EB count, our prototype is equipped with ElectroOculoGram (EOG) sensors, able to measure the EOG signal generated by the corneo-retinal standing potential between the front and the back of the human eye through simple electrodes installed on the glasses frame. A major advantage of this solution with respect to the state of the art lies in the fact that our implementation provides a completely transparent way to continuously monitor typical symptoms related to critical neurological diseases. This approach leverages the comfort and easiness of use inherent in wearing traditional glasses, while equipping the frame with useful sensors thus enabling

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TABLE I

A GENERAL OVERVIEW OF SOME OF THE MAIN STATE-OF-THE-ART WORKS. EB = EYE BLINK, EM = EYE MOVEMENTS, HMI = HUMAN MACHINE INTERFACE, EH = EHEALTH, BR = BASIC RESEARCH, SS = SAFETY & SECURITY, EOG = ELECTROOCULOGRAPHY, EEG = ELECTROENCEPHALOGRAPHY, IMV = IMAGE & VIDEO, RF = RADIO FREQUENCY, ML = MACHINE LEARNING, FA = FREQUENCY ANALYSIS, TA = TIME ANALYSIS, WVL = WAVELET, EMD = EMPIRICAL MODE DECOMPOSITION, TFA = TIME-FREQUENCY ANALYSIS, DE = DESKTOP EQUIPMENT, IOT = INTERNET OF THINGS

Refs	Observed Activity	Application Purpose	Exploited Signal	Employed Method	Employed Hardware	Brief Description
[15]	EB	HMI	EOG	ML	DE	This study aims to propose the voluntary eye blink detection method and apply it to the trigger switch of HMI systems. Accuracy around 97%.
[16]	EB	HMI, EH	RF	FA	IOT	Detection of EB using a Doppler sensor embedded inside a pair of glasses via spectrogram. No numerical results provided.
[17]	EM	HMI, EH	EOG, IMV	ML	IOT	Using EOG and IR-based signals authors discriminate between saccades, smooth pursuits, and vestibulo-ocular reflex movements. No accuracy provided.
[18]	EB	EH	IMV	ML	IOT	It employs an eigen-eye approach to detect EB in individual video frames. Accuracy around 96%
[19]	EB	BR	EOG	EMD (TA)	DE	It detects the EB through the EMD analysis. No accuracy provided.
[20]	EB	EH	IMV	ML	IOT	Detect EB from the normal side of the patient face and use them to stimulate the paralyzed side. Accuracy around 99%
[21]	EB	HMI	RF	ML	IOT	Detect the EB through wearable glasses exploiting proximity sensors. No accuracy provided.
[22]	EM	HMI	EOG	WVL (TFA)	DE	It describes a system for eye movements detection through EOG via microcontroller-based platform.
[23]	EM	BR	EOG	WVL (TFA)	DE	Wavelet-based method to denoise and detect eye movements from EOG signal.
[24]	EM	SS	EOG	ML	DE	Eye movements detection to control a robot for object tracking.
[25]	EM	BR	EEG	TFA	DE	Removal of eye movement and eye blink artifacts.
[26]	EB	BR	EEG	TA	DE	Detection and suppression of blink-artifacts in a single-channel EEG signal.
[27]	EB	SS	EOG, IMV	TA	DE	Evaluate the performance of an EOG and camera-based EB detection process in driving application.
[28]	EB	BR	EOG	TA	DE	Biometric authentication technique using EOG signals for eye blinking.
[29]	EM	EH	EOG	TA	IOT	Eye-tracking in Parkinson's patients through EOG.
[30]	EB	HMI	EOG	TA	DE	Real-time robot control through EOG within human-robot interaction.
[31]	EB	SS	EOG, IMV	TA	DE	Analysis of the relation between EOG and video analysis for EB detection and characterization of drowsiness in car crashes.

intelligent capabilities on this device. The main contributions of this work can be summarized as follows

1) we propose a novel pre-filtering scheme able to reduce noise in the EOG signal and to increase the EB counting accuracy with respect to the state-of-the-art solutions;
2) we provide an analytical representation of a matched filter able to efficiently identify EBs and consequently count their number;
3) we provide an extensive experimental campaign in order to validate the robustness of our approach with respect to the literature;
4) we compare the numerical results obtained with our prototype versus professional EOG equipment, showing the validity of our approach.

The rest of this paper is structured as follows: Section II provides a taxonomy of the related works available in literature and in Section III we present the considered scenario and the motivations leading to the proposed design. Section IV describes in detail the proposed eye blink detection algorithm and in Section V the performance results obtained in our extensive experimental analysis are provided. Finally, in the last Section we draw the conclusions.

II. RELATED WORK

The literature provides a significant amount of work tackling the problem of handling eye activity using smart devices. There are many solutions, each one belonging to different aspects of the state of the art as highlighted in Fig. 1. In the following we provide a taxonomy of the related works while a summary of the referenced papers is provided in Table I.

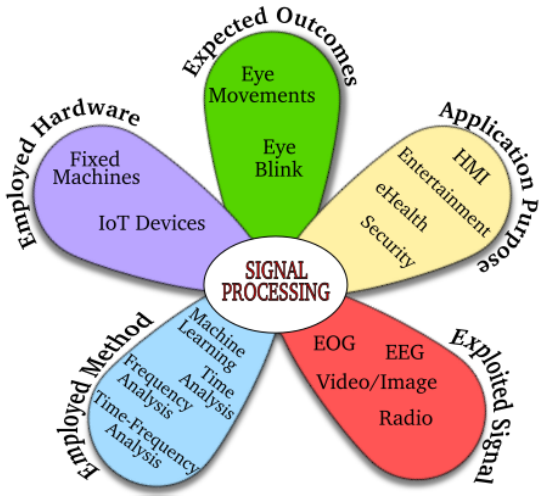

Fig. 1. Visual representation of the main aspects characterizing state-of-the-art solutions.

A. Expected Outcomes

When it comes to eye activity monitoring the two major expected outcomes are: eye blink (EBs) and eye movements. The former is related to the basic action of closing and opening the eyelids, which can be performed in normal or pathological ways while the latter refers to changing the direction of sight by moving the eyeballs in different positions. Since both actions are quite significant in interpreting the status of the

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eye, and of the subject in general, literature works tackle both the cases. Interesting papers such as [15], [21] are focused on EB detection and count, while other solutions prefer focusing on the direction of sight by monitoring movements [24], [25], [29] where EBs represent a nuisance component.

B. Application Purpose

The literature offers many solutions that can be categorized based on their application purpose. Indeed, eye movement monitoring can be exploited in many different scenarios, ranging from basic research [19], [28], to real-time eHealth [20], [29], to safety and security applications [24], [27], [31]. Other possible purposes for the employment of eye movement monitoring are related to the development of Human Machine Interfaces (HMIs) for real-time hands-free user interaction and robot control. In this framework, interesting contributions are presented in [15], [30], [32] and [33].

C. Exploited Signal

Another distinctive feature of the different state-of-the-art systems consists in the analyzed signal. Some works focus directly on the ElectroOculoGram (EOG) [28], [30], [31], while others infer the eye movement by exploiting the ElectroEncephaloGram (EEG) information [25], [26]. Also some solutions, such as [18], [20], are based on image/video analysis, providing visual observations on eye behaviour. Reference [16] is also an interesting work proposing radio frequency signal analysis to monitor eye movements through Doppler sensors.

D. Employed Method

Literature works report very different methods in terms of technical approaches used to solve the eye monitoring problem. In particular, a widely used technique is Time or Frequency Analysis of signals [26], [27]. This is the common approach used when we consider observations on EOG. Some of these methods rely on the use of Wavelet transforms [22], [23]. Other widely employed techniques are Machine Learning and Artificial Intelligence (AI), preferred in case of image/video based monitoring [17], [20], [21]. An alternative approach consists in the use of Empirical Mode Decomposition [19].

E. Employed Hardware

A fundamental differentiation among the literature works lies in the hardware employed to perform eye monitoring. There are two main approaches that can be identified in the state-of-the-art: *i)* the use of Desktop Equipment (DE) [22], [23] versus *ii)* wearable IoT devices [16]–[18]. Obviously there are several advantages and drawbacks that depend on the chosen hardware. These are related to the trade-off between the comfort, easiness of use and portability provided by wearable technology versus inevitable power and computational constraints caused by the mobile framework.

In the following we provide a detailed description of our proposed method, which focuses on EB detection and count in an IoT framework, using wearable smart glasses to collect EOG data.

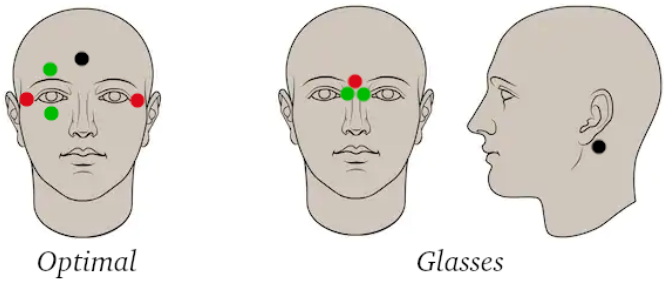

Fig. 2. The two configurations used in experimental tests for the position of EOG electrodes: *optimal* configuration (left) and *glasses* configuration (right).

III. The IoT Testing Scenario

A. General Introduction

IoT approach to eHealth is very different from traditional healthcare procedures due to its inherent characteristics. Namely, a professional medical equipment is usually very efficient and accurate but it is also very intrusive and uncomfortable for the patient. Moreover, a medical equipment usually consists in fixed desktop devices that are commonly used only inside hospitals or clinics. The great opportunity introduced by the IoT framework consists in exploiting user-friendly, easy-to-wear devices, provided with a technology able perform the same functions as traditional hardware, but with the added value of comfort, portability and transparency to the user.

In the following, we will compare the performances of our proposed EB detection solution versus a traditional medical approach, describing the main differences in terms of hardware and configuration. As will be shown later, systems that involve wearable IoT devices cannot confidently rely on traditional EOG signal processing techniques but on the contrary they need tailored solutions. For this reason, in this work we propose a specific and *ad-hoc* pre-filtering stage especially devised for EB detection in the wearable IoT framework.

B. Experimental Setup

1) Employed Hardware: The EOG is usually acquired using electrodes that are able to measure the signal generated by the corneo-retinal standing potential between the front and the back of the human eye. We acquired the EOG signal from a Desktop Equipment (DE) that aims at simulating a prototype of sensored wearable glasses able to detect and count EBs. Such prototype is shaped like a traditional pair of glasses, equipped with normal lenses, designed to be comfortably worn by anyone. They are also light and non-invasive.

A four-channel high-gain differential input amplifier and data acquisition system (MP35, Biopac Systems, Inc., USA) was used as DE together with Ag/AgCl electrodes placed at the positions shown in Fig. 2 to acquire the EOG signals. Signals were digitized at a sampling rate f_s equal to 100 Hz and transmitted to a PC via a USB interface.

2) Electrodes Configuration: To obtain the best possible signal quality the electrodes must be placed in specific positions on the patient's face. Unfortunately, this constraint cannot be granted by a wearable IoT device that trades the electrodes' position with the comfort of wearing simple glasses. In other words, the shape of the glasses forces specific positions for

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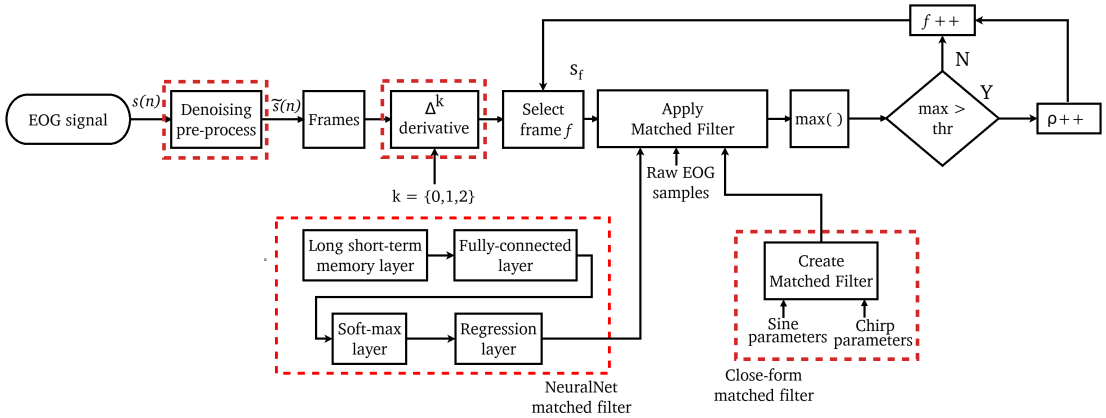

Fig. 3. General scheme of the proposed eye blink (EB) detection algorithm.

the electrodes which are different from the optimal ones. This great dissimilarity in hardware setup leads to the need of a different approach for EOG signal processing. In order to better understand the impact of the EOG electrodes positioning, in the following we will consider two different configurations for the EOG signal acquisition:

- *Optimal*: it is the same electrodes positioning related to professional medical desktop equipment. It is the best electrodes positioning.
- *Glasses*: the electrodes position follows the physical contact points of the glasses with the face.

Figure 2 shows a graphical representation of both the configurations used in the experimental tests where the red dots represent the horizontal channel, the green dots refer to the vertical one and the black dots correspond to ground.

In the *optimal* configuration, the electrodes are put above and beneath one eye, and at the external corner of both eyes, while ground is measured on the forehead. This configuration is shown in Figure 2 (left part) In the literature, many papers acquire the EOG signal by using this configuration [19], [28]. Although this configuration achieves minimal perturbation on the measured signals, it is a very uncomfortable setup for the patient due to the intrusiveness of the solution. Moreover, the electrodes position in this approach cannot be granted by using wearable glasses as EOG sensing device since the electrodes must be put where the glasses frame has contact with the patient's skin. These specific locations are the glasses bridge, located at the top of the nose, and the temples, where reference ground is measured. This configuration is shown in Figure 2 (right part). Since the *optimal* configuration provides the best results in terms of performance, in the present work we run experimental tests by using the *optimal* EOG configuration as a benchmark for the *glasses* one.

3) Test Conditions: To cover all the possible scenarios we have considered three different conditions:

- Staring-Eyes (SE) - the subject is looking fixed right in front of him and blinks naturally. No other actions were performed (no head movements, no speaking);
- MoVement (MV) - the patient can move his eyes and his head as he wants while he keeps on blinking naturally. No speaking allowed.
- SpeaKing (SK) - the subject can speak throughout all the acquisition and blinks naturally. He cannot move his eyes and his head.

IV. The Proposed Eye Blink (EB) Detection Algorithm

The overall Eye Blink (EB) detection algorithm is sketched in Fig. 3. The blocks surrounded by the red dashed line highlight the main contributions of this paper.

A. Denoising Pre-Processing Filter

It is the first block that is applied to the raw EOG signal. It aims at removing the noise and to polish the EOG signal in order to prepare the signal for further processing. In this paper we propose a novel denoising pre-processing filter and we compare its performance with one of the most common state-of-the-art denoising technique used for EOG signals which is based on the use of wavelet transform. In the following, both the filtering methods are described.

1) Wavelet Denoising: The employment of wavelet transform is a very powerful tool for signals such as EOG, which are corrupted by noise and highly non-stationary by nature [23]. A wavelet is a small wave which oscillates and decays in the time domain. Unlike the Fourier transform, wavelets can have infinite varieties which are fundamentally different from each other. The ones which have a finite time extent are known as discrete wavelets, otherwise they go by the name of continuous wavelets. Nowadays the wavelet analysis is used for processing and synthesis of non-stationary signals, solving problems of compressing and coding information, image processing and practice of pattern recognition for medical signals [34]. As suggested by [23], [28], [35], [36] and many other references therein, along with the Haar and Symlet wavelets, the Daubechies wavelet seems the one providing the best results when it comes to biomedical signal processing. Consequently, following the approach reported in the aforementioned papers, in this work we employed Daubechies 4 (also called *Dab4*) as the mother wavelet for the denoising process. The classical wavelet denoising scheme is sketched in Fig. 5.

2) Proposed Denoising Filtering: As an alternative to Wavelet-based denoising, in the following we propose a novel EOG signal denoising scheme. The scheme describing this

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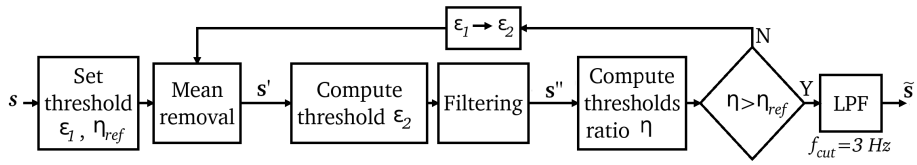

Fig. 4. The proposed iterative filtering (IF) scheme for the EOG signal.

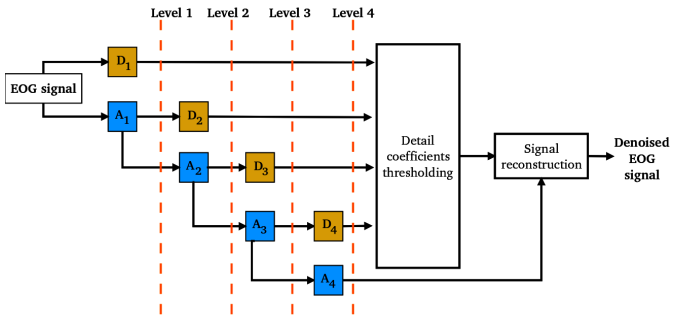

Fig. 5. The classical wavelet denoising scheme.

method is provided in Fig. 4 where $\mathbf{s}$ is the vectorial notation for the EOG signal $s(n)$, $n \in [0, N-1]$.

As a preliminary action it is necessary setting two thresholds ϵ_1 and η_{ref}: this is done by empirically initializing $\epsilon_1 = 10^{24}$ and $\eta_{ref} = 0.95$. The next step consists in removing the mean from the signal $s(n)$ thus obtaining a new signal $s'(n) = s(n) - \mu_s$ where $\mu_s = \frac{1}{N}\sum_{n=0}^{N-1} s'(n)$. The threshold ϵ_2 is then computed as $\epsilon_2 = \alpha \cdot \sigma_{s'}$ where $\sigma_{s'} = \sqrt{\frac{\sum_{n=0}^{N-1}[s'(n)]^2}{N}}$ is the standard deviation. The quantity α is a smoothing factor (empirically set to 0.5) that trades the severity of the threshold operation. $s'(n)$ is then filtered through a non-linear, soft-threshold block to obtain the signal $s''(n)$ as follows:

$$s''(n) = \begin{cases} \frac{1}{2T}\sum_{t=1}^{T} s'(n-t) + s'(n+t), & \text{if } |s'(n) \geq \epsilon_2| \\ s'(n) - \epsilon_2, & \text{if } |s'(n) < \epsilon_2| \end{cases} \tag{1}$$

$\forall n \in [0, N-1]$. The quantity T is the filter span, set to 5 in this paper. Finally, the ratio $\eta = \frac{\epsilon_2}{\epsilon_1}$ is computed and compared with its reference threshold η_{ref}: if $\eta \leq \eta_{ref}$ then ϵ_1 is updated by setting its value to ϵ_2. Vice versa if $\eta > \eta_{ref}$ the pre-process stage concludes by applying a LPF with cutting frequency set to 3 [Hz], thus obtaining as output the pre-processed signal $\tilde{s}(n)$.

B. Eye Blink (EB) Detection

The Eye Blink (EB) detection is performed by dividing into frames the filtered and denoised EOG signal $\hat{s}(n)$ by using the scheme sketched in Fig. 3. Each frame presents an overlap percentage, denoted with O, with the following one and it is composed of 60 samples (equivalent to 0.6 seconds). The next block is the k-derivative, $k \in \{0, 1, 2\}$, where k is the derivation order. The employment of such a block is motivated by the fact that the derivation allows enhancing the abrupt changes in the signal and it provides a more identifiable waveform for the Matched Filter (MF). However, at the same time, derivation may also introduce some additional noise and therefore the derivation order k must be carefully chosen. Finally, to each frame a matched filter is then applied and the maximum value is extracted. Similarly to [37], such maximum is compared with an empirically determined threshold *thr* which allows to determine or not the presence of an EB.

1) Matched Filter: It is a well-known signal processing technique for pattern recognition which has been widely used in several fields such as image processing and eHealth applications. Methods based on MF provide a distance representing the similarity between a test signal and a predefined waveform (template). MF has been previously used in detecting blink events in single-channel EEG/EOG applications (see [26] and references therein). The major concerns of this method are: *i)* the definition of the template and *ii)* the definition of the threshold. Both these concerns can be easily solved if the analytical expression of the considered signal is known a-priori which is a condition usually granted in classical telecommunication systems. Unfortunately, this is something that is not true when it comes to biomedical and eHealth systems. Indeed, to the best of the authors' knowledge, no close-form expression is available in the literature that can describe the EB signal. Consequently, one of the main contributions of this paper consists in proposing and comparing different approaches to realize the MF:

1) directly from the EOG samples, Average Matched Filter (AvgMF);
2) estimating it from the EOG samples a Neural Network used as regression model (NN MF);
3) by providing a close-form through: *i)* sum of sines (Sine MF) model and *ii)* Chirp (Chirp MF) model;

In the following each one of the former approaches will be deeply analyzed.

a) Average matched filter: In order to design the AvgMF we have recorded EOG acquisitions of about $30s$ from 5 subjects, sampling at $f_s = 100$ [Hz]. Among these acquisitions we have manually isolated 5 EBs for each subject, thus obtaining $B = 25$ different blinks $\tilde{s}_1(n), \tilde{s}_2(n), \cdots \tilde{s}_b(n), \cdots \tilde{s}_B(n)$, where $\tilde{s}_b(n)$ is a pre-processed EOG sequence containing just one blink and $b \in [1, B]$ identifies the $b-th$ blink. The obtained EB segments were manually aligned and then the sequence $\overline{s}(n)$ is computed by averaging the elements $\tilde{s}_b(n)$.

An in depth analysis of the proposed approach showed that the use of the pre-processed signal might not be sufficient to effectively characterize the EB waveform, thus suggesting to explore the use of the first and second-order derivatives of the EOG signal. Therefore, we investigated the performance obtained with all the three aforementioned approaches. In particular, the reason for the second-order derivative is two-fold: *i)* it allows enhancing the abrupt changes in the signal and *ii)* it provides a more identifiable waveform for the matched

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filter $m(n)$. Indeed, the use of this specific approach is crucial in obtaining a more accurate identification of EBs.

As a result, the sequence $\overline{s}(n)$ obtained by employing the second order derivative of the EOG signal can be expressed as:

$$\overline{s}(n) = \frac{1}{B}\sum_{b=1}^{B} \Delta^2\left[\tilde{s}_b(n)\right] \tag{2}$$

where Δ^2 denotes the second-order discrete derivative. Finally, we define the AvgMF $m_{AVG}(n) = \overline{s}(Q-1-n)$, $\forall n \in [0, Q-1]$, where Q is the length of the sequence $\overline{s}(n)$.

b) Sine matched filter: In this approach we model the MF $m(n)$ as a Summation Of K Sinusoidal components (SOS) with different amplitude, frequency, phase and offset. The employed mathematical expression is the following:

$$m_{SOS}(\boldsymbol{\theta}, n) = \sum_{k=1}^{K} A_k \cdot sin(2\pi f_k \mathbf{n}_k + \phi_k) - O_k \tag{3}$$

where we have set $K = 3$. The vector $\mathbf{n}_k$ contains the values of the independent variable for each sinusoidal component, while $n \in [0, N-1]$ is the independent variable describing the obtained expression $m_{SOS}(n)$ of the matched filter. The quantity N is the number of samples composing the matched filter that can be obtained given the average EB duration L as $N = \lfloor \frac{L}{f_s} \rceil$, where the symbol $\lfloor \cdot \rceil$ is the rounding operator. Considering the state of the art in the field (see [15], [17], [28]) and the measurements made in our experiments, we have decided to set $L = 0.55$ seconds.

Finally, the SOS parameter matrix $\boldsymbol{\theta}$ is defined as

$$\boldsymbol{\theta} = \left([\mathbf{A}]^T[\mathbf{f}]^T[\boldsymbol{\phi}]^T[\mathbf{O}]^T\right) = \begin{pmatrix} A_1 & f_1 & \phi_1 & O_1 \\ A_2 & f_2 & \phi_2 & O_2 \\ A_3 & f_3 & \phi_3 & O_3 \end{pmatrix} \tag{4}$$

where its elements are amplitudes, frequencies, phases and offsets of each sinusoidal component. Starting from the average EB $\overline{s}(n)$ reported in Eq. 2, we provide a simple algebraic method for estimating the elements of the matrix $\boldsymbol{\theta}$.

We begin by defining the pairs $< l_k, m_k >$, $k \in [1, K]$ as follows:

$$\begin{cases} l_1 = \underset{n\in[0,N-1]}{\arg\max}\ \overline{s}(n); & m_1 = \underset{n\in[0,N-1]}{\max}\ \overline{s}(n) \\ l_2 = \underset{n\in[0,N-1]}{\arg\min}\ \overline{s}(n); & m_2 = \underset{n\in[0,N-1]}{\min}\ \overline{s}(n) \\ l_3 = \underset{n\in[\lfloor\frac{N}{2}\rceil,N-1]}{\arg\max}\ \overline{s}(n); & m_3 = \underset{n\in[\lfloor\frac{N}{2}\rceil,N-1]}{\max}\ \overline{s}(n) \end{cases} \tag{5}$$

The graphical meaning of each quantity is reported in Fig. 6, where a sample representation of the second-order derivative of the EB waveform is provided.

The vector $\mathbf{n}_k$ can be defined starting from l_1, l_2 and l_3 as $\mathbf{n}_k = \left[[0, l_2]^T, [l_2+1, l_3]^T, [l_3+1, L]^T\right]$.

The offsets estimation is carried out by using some graphical consideration. Without losing any validity, in the proposed approach we set the estimation for O_1 and O_3 equal to zero. This choice is also motivated from a visual inspection of the average EB $\overline{s}(n)$. Consequently, the only offset needed to be estimated is O_2 that can be achieved as $O_2 = A_2 - m_3 = m_2 - m_3$.

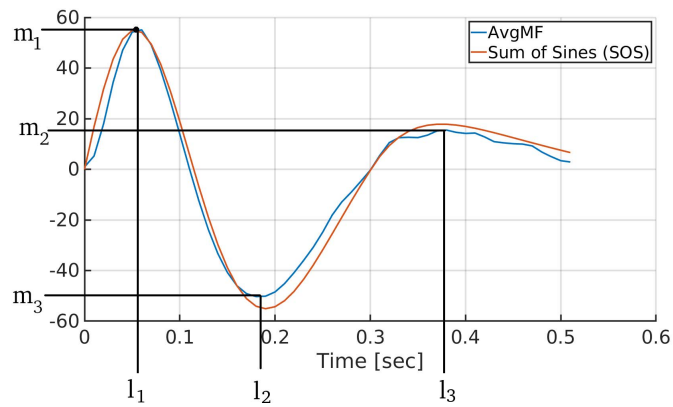

Fig. 6. Example of the second-order derivative of the EB signal and its approximation using the proposed SOS method.

Given the quantities obtained from Eq. 5 we can estimate the frequencies and amplitudes as follows:

$$\begin{cases} f_1 = 2\dfrac{l_2 - l_1}{f_s}; & A_1 = m_1 \\ f_2 = 2\dfrac{l_3 - l_2}{f_s}; & A_2 = m_2 \\ f_3 = 2\dfrac{L - l_3}{f_s}; & A_3 = m_3 \end{cases} \tag{6}$$

The phase estimation is carried out by firstly imposing $\phi_1 = 0$ as a reasonable starting point and then by constraining a smooth transition between each sinusoidal component. To guarantee that such condition holds, for the first two sinusoidal components we should impose that:

$$sin(2\pi f_1 n + \phi_1)|_{n=l_2} = sin(2\pi f_2 n + \phi_2)|_{n=l_2}. \tag{7}$$

Considering the definition of f_1 and f_2 reported in Eq. 6 and the phase unwrapping between $[-\pi, \pi]$, this yields to:

$$\phi_2 = \frac{3}{2}\pi(1 - \frac{f_2}{f_1}) \tag{8}$$

An analogous condition must be imposed in order to estimate the phase of the third sinusoidal component:

$$sin(2\pi f_2 n + \phi_2)|_{n=l_3} = sin(2\pi f_3 n + \phi_3)|_{n=l_3} \tag{9}$$

Provided the same considerations as above, we obtain:

$$\phi_3 = \frac{1}{4} \cdot (\frac{3}{f_1} + \frac{2}{f_2} - \frac{5}{f_3}) \tag{10}$$

In conclusion, the overall SOS parameter matrix $\boldsymbol{\theta}$ can be written as:

$$\boldsymbol{\theta} = \begin{pmatrix} m_1 & 2\dfrac{l_2 - l_1}{f_s} & 0 & 0 \\ m_2 & 2\dfrac{l_3 - l_2}{f_s} & \dfrac{3}{2}\pi(1 - \dfrac{f_2}{f_1}) & m_2 - m_3 \\ m_3 & 2\dfrac{L - l_3}{f_s} & \dfrac{1}{4} \cdot (\dfrac{3}{f_1} + \dfrac{2}{f_2} - \dfrac{5}{f_3}) & 0 \end{pmatrix} \tag{11}$$

c) Chirp matched filter: Similarly to the previous case, we provide another analytical model for the EB based on the quadratic down ChiRP (CRP). The rationale behind this approach is that the EB waveform can be represented as a single sinusoidal component where the frequency decreases

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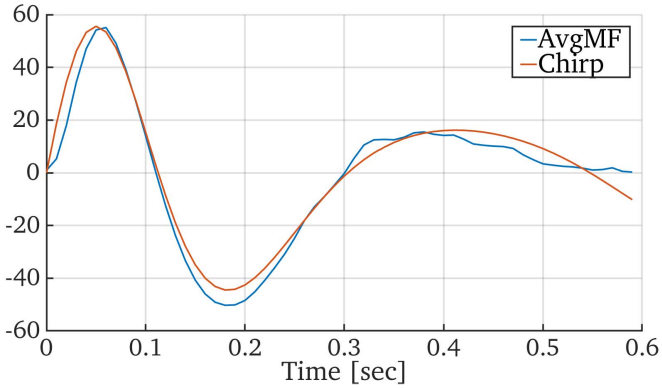

Fig. 7. Example of the second-order derivative of the EB signal and its approximation using the proposed chirp method.

with time. Formally, the mathematical expression for the chirp matched filter $m_{CRP}(n)$ is the following:

$$m_{CRP}(\boldsymbol{\lambda}, n) = C \cdot sin(2\pi\nu n - \alpha n^2), \quad n \in [0, N-1] \tag{12}$$

where $\boldsymbol{\lambda}$ is the chirp parameter vector defined as $\boldsymbol{\lambda} = \left[C, \nu, \alpha\right]$. The quantities ν and α are the base chirp frequency and the frequency shift, respectively.

Although many papers in the literature have tackled the problem of estimating the chirp parameters (see [38], [39] and references therein), they always assume that the noisy signal from which to infer the parameter estimation is an ideal chirp signal corrupted by noise. In our problem, this hypothesis does not hold since the second-order derivative of the EB waveform is a natural and biological signal, whose actual analytical expression is unknown. Consequently, the algorithms reported in the aforementioned papers are not suited for the scenario considered in this work. Therefore, starting from the results reported in [40] we compute the values of $\boldsymbol{\lambda}$ through a Least-Squared Estimation (LSE) approach. Formally:

$$\boldsymbol{\lambda}^* = \min_{\boldsymbol{\lambda}} \sum_{n=0}^{N-1} \left[\overline{s}(n) - m_{CRP}(\boldsymbol{\lambda}, n)\right]^2 \tag{13}$$

Fig. 7 reports an example of the second-order derivative of the EB signal and its approximation using the proposed Chirp method.

Finally, Table II reports the numerical values obtained by employing the two closed-form approaches for the MF: *i)* Sum Of Sines (SOS) and ChiRP (CRP) along with the Root Mean Square Error (RMSE) from the average MF. As it is possible to note from the Table, both the RMSE are very low and similar: the SOS approach provides the lowest RMSE (equal to 2.48) while the CRP shows a RMSE of 5.12.

TABLE II

NUMERICAL VALUES OBTAINED FOR THE TWO CLOSED-FORM APPROACHES FOR THE MF: *i)* SUM OF SINES (SOS) AND CHIRP (CRP) ALONG WITH THE ROOT MEAN SQUARE ERROR (RMSE) FROM THE AVERAGE MF

Sum Of Sines (SOS)											
A			**f**			ϕ			**O**		
A_1	A_2	A_3	f_1	f_2	f_3	ϕ_1	ϕ_2	ϕ_3	O_1	O_2	O_3
50.2	35.6	2.3	4.5	2.4	1.7	10^{-6}	1.5	10^{-6}	-0.08	2.3	3.1
RMSE: 2.48											
ChiRP (CRP)											
C			ν						α		
60.44			6.01						1.64		
RMSE: 5.12											

V. EXPERIMENTAL RESULTS

In this section we will present the experimental results obtained in our performance analysis. As described in Section III, in the present work we consider two different scenarios depending on the electrodes setup: *optimal* and *glasses*. We run experimental tests using the *optimal* EOG electrodes configuration as a reference performance with respect to the *glasses* setup, since the *optimal* configuration should provide the best results in terms of performance. Moreover, to prove the effectiveness of the proposed approach we have tested our solution considering different scenarios. Furthermore, to obtain a fair and comprehensive comparison, we have compared our approach also with another method based on Empirical Mode Decomposition (EMD) [19]. It is important to notice that experimental conditions are quite realistic, since during the tests subjects were free to move their eyes and speak (MV + SK conditions) thus adding more noise to the acquired signal.

A. Glasses Configuration

In a first round of experiments, we compared the results of EB detection in the case where the electrodes for measuring the EOG signal are applied to the subject in the *glasses* configuration. We define the error as the difference, in absolute value, between the actual number of performed EBs and the estimated ones. Fig. 8 shows the Cumulative Density Function (CDF) of the EB error obtained when performing EB detection with different algorithms. We ran a performance analysis comparing the AvgMF, SOS, CRP and NN approaches versus the EMD algorithm. We have also analyzed the impact of different EOG filtering schemes (classical Wavelet-based denoising approach versus the proposed pre-processing Iterative Filtering) and derivatives (no signal derivative, first-order derivative and second-order derivative). The proposed AvgMF is able to perform better than other methods such as EMD in all considered cases. Furthermore, the proposed pre-processing scheme is very effective in cleaning the signal especially when derivatives are applied. On the contrary, the Wavelet-based approach proves more useful when we consider directly the pre-processed signal for eye blink detection. Also, SOS, CRP and NN methods can outperform the EMD approach independently of the considered pre-filtering scheme.

B. Optimal Configuration

The second round of experiments is related to EB detection errors in the case where the electrodes for measuring the EOG signal are applied to the subject in the *optimal* configuration. Fig. 9 shows the comparison of the results obtained using the same plots described in the former scenario. The main difference is related to the type of signal acquired by electrodes, which are placed in the position required for the professional medical EOG acquisition. Also in this configuration results confirm that the proposed pre-processing filter provides better performance when derivatives are applied to the EOG signal.

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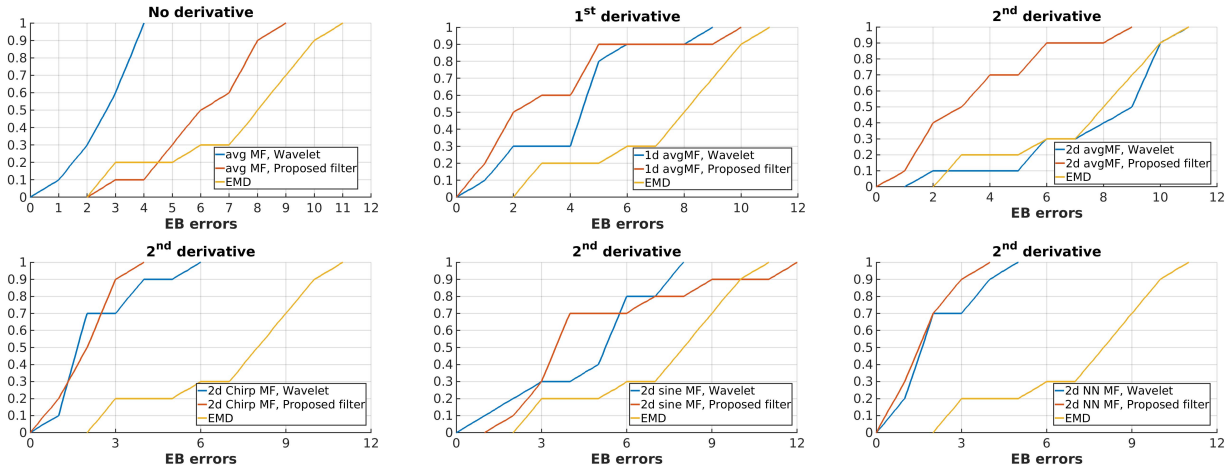

Fig. 8. Cumulative Density Functions (CDFs) for the EB error for the case where the electrodes are put in the *glasses* configuration. During these tests the users can talk and move their heads freely to simulate a real-case scenario.

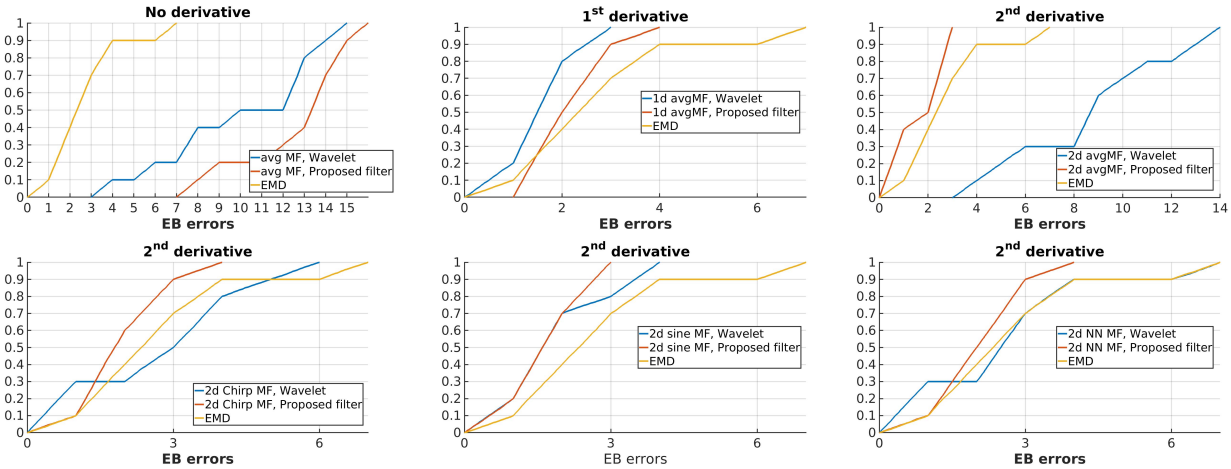

Fig. 9. Cumulative density functions (CDFs) for the EB error for the case where the electrodes are put in the *optimal* configuration.

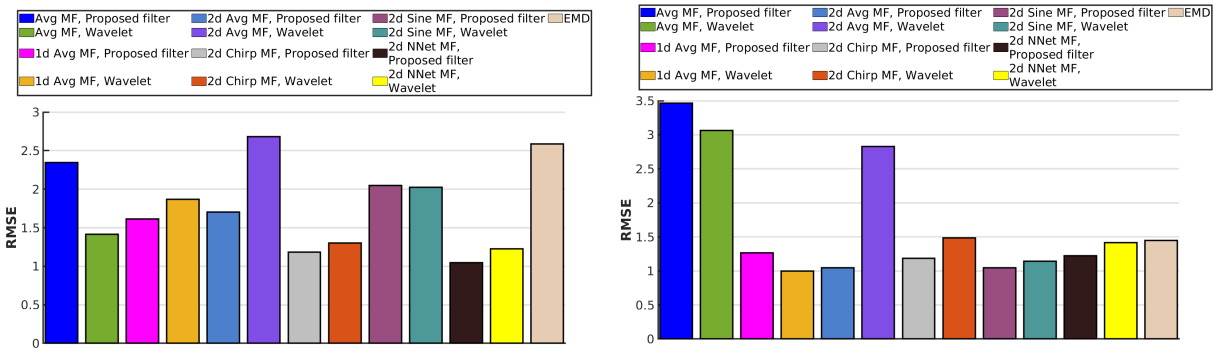

Fig. 10. Bar plot of the RMSE of EB count in *glasses* configuration (left) and optimal configuration (right) for the considered EB counting algorithms.

It can be noted that the only case in which the EMD approach performs better than other methods is when we consider directly the raw signal for EB detection. Finally, the most accurate results are obtained by employing the AvgMF or SOS filter to the second-order derivative of the EOG signal, when pre-processed with our proposed filter.

C. Overall Comparison

The last round of experiments is related to the overall comparison of different EB counting algorithms in terms of RMSE. Fig. 10 shows the performance obtained in the *glasses* configuration (left) and *optimal* configuration (right) scenarios. Almost all considered approaches provide comparable overall results in terms of EB count error and, as expected, RMSE values are slightly lower in the *optimal* scenario for the majority of the approaches. Anyway, also the *glasses* configuration can provide good results even if the electrodes position is sub-optimal, thus providing a valuable trade-off between accuracy and comfort. Furthermore, we can notice that some methods such as AvgMF with Wavelet-based filtering, SOS MF and

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TABLE III

OVERALL COMPARISON OF THE RMSE FOR THE DIFFERENT APPROACHES FOR THE TWO CONSIDERED CONFIGURATIONS: *i) Optimal* AND *ii) Glasses*. AVG = AVERAGE, MF = MATCHED FILTER, WVL = WAVELET, IF = ITERATIVE FILTER (SEE FIG. 4), NN = NEURAL NETWORK, EMD = EMPIRICAL MODE DECOMPOSITION

	***Optimal* configuration**												
	Δ^0 Avg MF WVL	Δ^0 Avg MF IF	Δ^1 Avg MF WVL	Δ^1 Avg MF IF	Δ^2 Avg MF WVL	Δ^2 Avg MF IF	Δ^2 Chirp MF WVL	Δ^2 Chirp MF IF	Δ^2 Sine MF WVL	Δ^2 Sine MF IF	Δ^2 NN MF WVL	Δ^2 NN MF IF	EMD
Fixed	5.13	6.84	2.51	7.02	2.91	3.05	2.26	2.37	2.61	2.24	2.25	2.22	4.42
Spk	5.59	3.89	4.29	4.29	5.29	6.19	7.09	5.39	6.00	5.39	7.22	6.29	6.20
Mov	5.09	3.49	1.50	1.98	5.42	1.44	3.30	2.01	2.23	1.39	4.5	1.99	4.39
Spk+Mov	7.54	5.42	5.39	4.90	4.79	3.19	4.09.	5.03	7.55	3.39	9.31	7.89	9.32
	***Glasses* configuration**												
	Δ^0 Avg MF WVL	Δ^0 Avg MF IF	Δ^1 Avg MF WVL	Δ^1 Avg MF IF	Δ^2 Avg MF WVL	Δ^2 Avg MF IF	Δ^2 Chirp MF WVL	Δ^2 Chirp MF IF	Δ^2 Sine MF WVL	Δ^2 Sine MF IF	Δ^2 NN MF WVL	Δ^2 NN MF IF	EMD
Fixed	8.49	7.99	3.10	3.80	2.99	1.09	3.56	2.09	3.04	2.99	3.41	3.47	2.19
Spk	6.23	4.56	5.27	4.03	4.36	5.12	6.35	4.12	5.03	4.99	4.33	4.95	5.47
Mov	5.75	4.12	2.23	2.96	5.28	2.35	3.36	2.55	2.78	2.99	3.98	3.47	5.01
Spk+Mov	6.39	6.47	5.66	4.89	5.57	5.08	4.36	4.18	5.65	4.97	4.03	3.99	8.79

EMD perform differently in the two scenarios. Globally, the best results are obtained in the case of AvgMF applied to the first derivative of EOG signal with Wavelet-based filtering when considering the optimal configuration. In the *glasses* scenario, we can see that the lowest RMSE is provided by the NN and CRP MF applied to the second-order derivative of the EOG signal, independently of the considered pre-processing filter. Same considerations can be inferred from Table III which reports a comparison of the RMSE for all the possible approaches for both the *optimal* and *glasses* configurations.

VI. CONCLUSION & FUTURE WORK

In this paper we analyzed the framework of IoT with respect to its applications in medical sciences and healthcare. In particular, we tackled the problem of the early detection of neurological symptoms by leveraging IoT wearable technology to perform in-home monitoring in a transparent way with respect to the final user. We carried out an extensive experimental campaign, comparing the performance of our system with other existing solutions. Results show that the employed solutions are able to efficiently perform EBs detection, achieving comparable accuracy with respect to professional medical desktop equipment, and providing to the patients at the same time the benefit of portability, comfort and easiness of use. Future research on this topic will involve a deeper analysis of the EOG noise in order to realize a more robust and efficient MF by taking into account the different typologies of electrodes that the EOG machines use (i.e., wet or dry).

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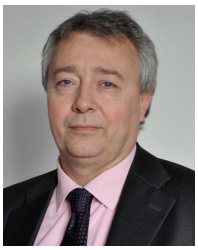

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